



CLASS PARTICIPATION COVERSHEET

Unit of Study: **MIT653 Research Methods**

Assignment Title: **Class Participation – Week 01**

Trimester and Year: **Trimester 2, 2026**

Lecturer Name: **Dr. Sudath Heiyanthuduwege**

DECLARATION

The work contained in this assignment is my own work and has not been copied from other sources or been previously submitted, unless otherwise specifically acknowledged.

I understand that failure to comply can lead to severe penalties.

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Student Name: Salman Ashraf

Date Submitted: ____dd____mm____yy

Signed: _____ Date: ____dd____mm____yy

EXAMINER ONLY

Due Date: ____dd____mm____yy

MARK: _____

FINAL MARK: _____

Section A: Research Process

Case Study: Exploring the Relationship between Exercise and Academic Performance

1. What is the relationship between exercise and academic performance among college students?

This question came from a gap in the literature review. Most studies already looked at school kids and teenagers and found a positive link between exercise and grades. But not many studies looked at college students. College students have different problems like more study pressure, stress, and managing their own time. So the researcher picked this group because it was not studied much before.

2. How was the hypothesis developed based on the research question and literature review?

The hypothesis is: "Regular exercise positively affects academic performance among college students." It was made using the idea already found in past research, that exercise usually helps grades. The researcher just took that same idea and applied it to college students, since that was the group chosen in the research question.

3. What type of research design was used in this case study?

The study uses a quantitative research design. It is basically a survey type study done at one point in time (not over months or years), and the data collected is turned into numbers so it can be tested with statistics.

4. What data collection methods were employed to test the hypothesis?

A survey was given to college students at one college. The survey asked about their exercise habits, their grades/academic performance, their stress levels, and some basic personal information (like age, gender, etc). After collecting this, the researcher used descriptive statistics, regression analysis, and hypothesis testing to check if exercise and academic performance are really connected.

5. Were there any unexpected results or limitations in the study? How were they addressed?

The case study only talks about the plan for the research, it does not actually show real results, so there are no results to check as "unexpected" yet. But some limitations can still be pointed out:

- Only one college was used, so the results might not apply to all colleges.
- Since it is a survey, students might not answer honestly about their exercise or stress levels.
- Even if a link is found, it does not prove exercise directly causes better grades, other things could be involved too.

These limitations can be handled by mentioning them at the end of the study and suggesting that future research use more colleges and better methods.

6. How does this study contribute to the existing knowledge in the field?

This study adds new information about a group (college students) that was not studied much before. It can also help find other things, like stress or time management, that might affect how exercise and grades are connected.

7. What are some potential implications or applications of the study's findings?

If the study shows that exercise really helps grades, colleges could use this information to encourage students to exercise more, like adding gym time or wellness programs. It is also a cheap and easy way to help students do better in school without spending a lot of money.

8. Are there any ethical considerations that need to be considered in this study?

Yes, a few things need to be kept in mind:

- Students should agree to take part in the survey on their own (informed consent).
- Their answers, especially about grades and stress, should be kept private.
- No one should be forced to join, and they should be able to leave the study anytime.
- The study should probably get approval from the college before starting.

9. How could this study be further expanded or improved in future research?

The study could be made better by:

- Using students from more than one college.
- Doing the study over a longer period of time instead of just once.
- Using something like a fitness tracker instead of just asking students how much they exercise.
- Talking to students directly (interviews) to understand things better.

10. What are the key takeaways from this case study for conducting research with research questions and hypotheses?

The main takeaway is that a good research question comes from finding a gap in what has already been studied. The hypothesis should directly connect to that question and be based on what past research already suggests. Also, the way the data is collected should match what the hypothesis is trying to test, and researchers should think about limitations and ethics before starting the study, not after.

Section B: Characteristics of a Good Researcher

Matching Descriptions and Self-Rating Table:

Description	Characteristic	Your Rating (1-10)
Constantly seeking and exploring new knowledge and discoveries	Curious / Inquisitive	8/10
Carefully evaluating risks and making sensible, informed decisions	Prudent / Careful Decision-Maker	8.5/10
Persisting towards goals even when facing setbacks or difficulties	Persistent / Resilient	7/10
Planning ahead, anticipating problems, and identifying future research directions	Forward-Thinking / Visionary	7/10
Being trustworthy in how you use resources, conduct research, and report results	Honest / Trustworthy (Integrity)	8/10

Description	Characteristic	Your Rating (1-10)
Thinking critically and systematically to analyze, interpret, and evaluate information	Analytical / Critical Thinker	9/10
Demonstrating passion and deep commitment to your research work	Passionate / Committed	9/10
Communicating effectively with different stakeholders through writing and speaking	Effective Communicator	7/10
Thinking boldly and creatively to push the boundaries of knowledge	Creative / Innovative	7.5/10
Adapting to change and being flexible when facing new challenges or evidence	Adaptable / Flexible	8/10
Total Score (out of 100):		79/100